FIFA World Cup Match Outcome Prediction Using Machine Learning Techniques

Advanced Python Programming -- Group F

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1. Introduction

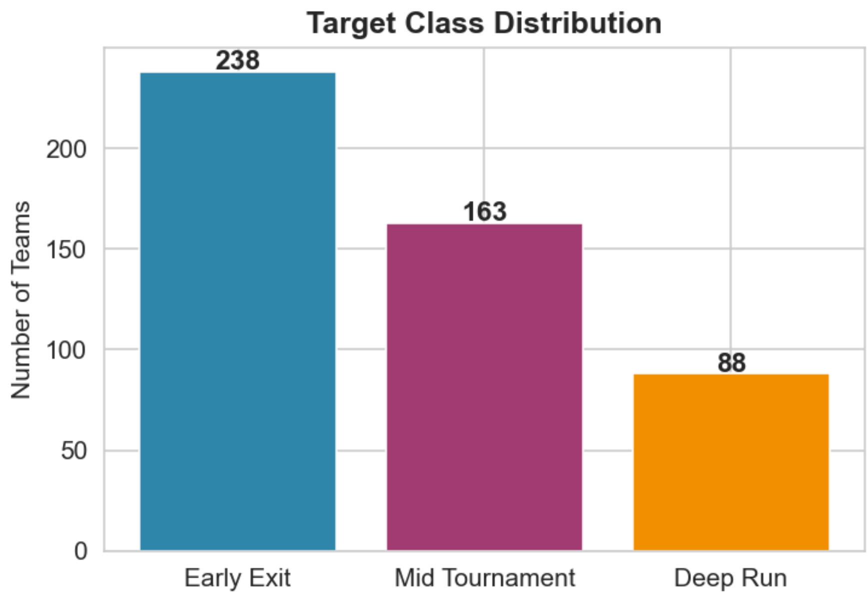
This project applies supervised machine learning techniques to predict FIFA World Cup match outcomes. Unlike traditional match-level prediction (win/draw/loss), this study predicts tournament stage progression for national teams based on historical performance data spanning from 1930 to 2022. The target variable is a three-class categorisation: Early Exit (group stage), Mid Tournament (round of 16 / quarter-finals), and Deep Run (semi-finals or better).

Three models are compared: Logistic Regression (baseline), Random Forest, and XGBoost. All features are engineered chronologically -- only information available before each tournament is used -- to prevent data leakage and simulate realistic prediction conditions.

2. Data Overview

The training dataset contains 489 entries, each representing a national team's participation in a specific FIFA World Cup tournament (1930-2022). The test dataset contains 48 teams slated for the 2026 FIFA World Cup. Each training record includes: country, confederation, region, tournament year, matches played, total goals scored, and the stage reached.

Target classes after consolidation: Early Exit (n=238), Mid Tournament (n=163), Deep Run (n=88).



3. Feature Engineering

Eight features were engineered, all computed using only data from tournaments preceding the one being predicted. This chronological constraint ensures no future information leaks into the training process.

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1. Previous Appearances -- number of prior World Cup participations
2. Best Previous Stage -- highest stage ever reached (encoded 0-2)
3. Avg Goals/Match (Historical) -- mean goals per game across all past tournaments
4. Years Since Debut -- time elapsed since first World Cup appearance
5. Last Appearance Stage -- stage reached in the most recent prior tournament
6. Confederation -- encoded label for continental federation (UEFA, CONMEBOL, etc.)
7. Region -- encoded label for geographic region
8. Year (normalised) -- tournament year scaled to [0, 1]
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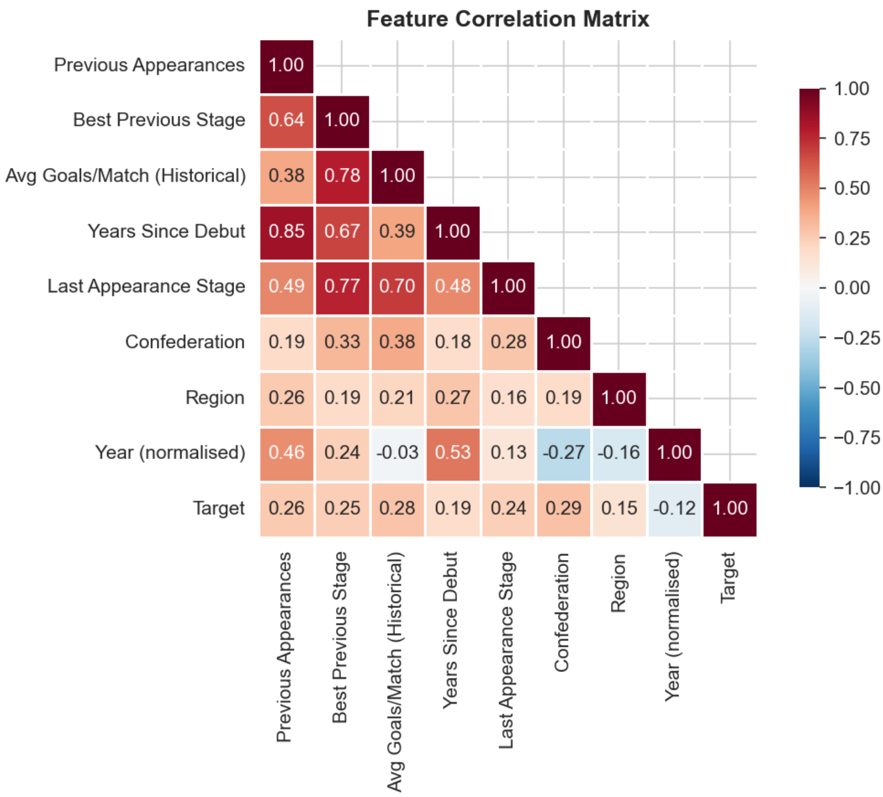


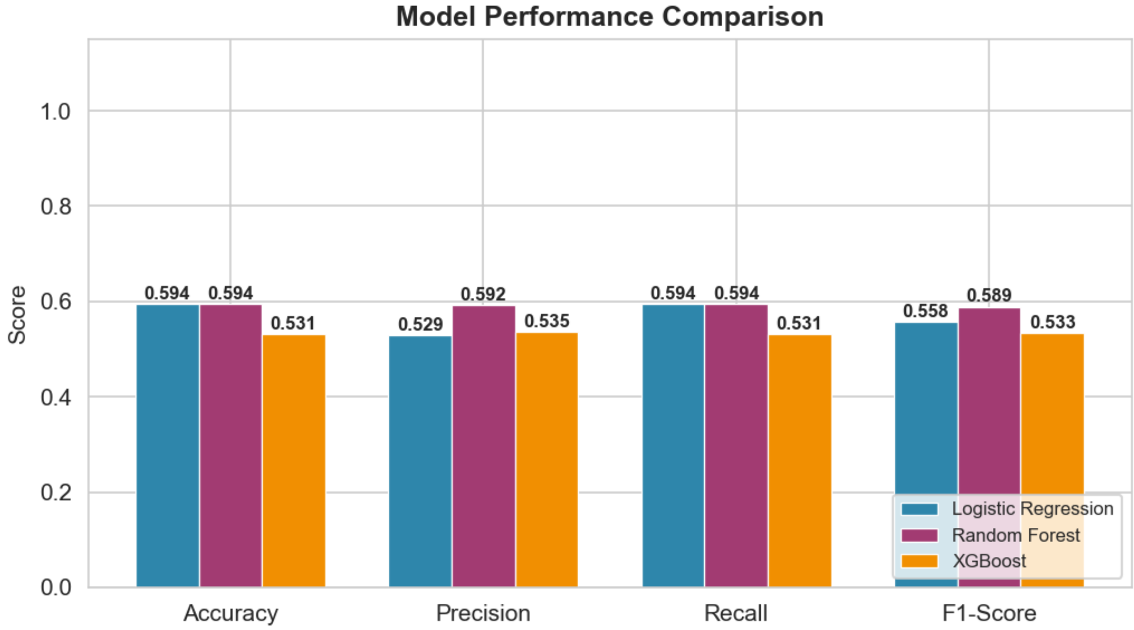
4. Methodology

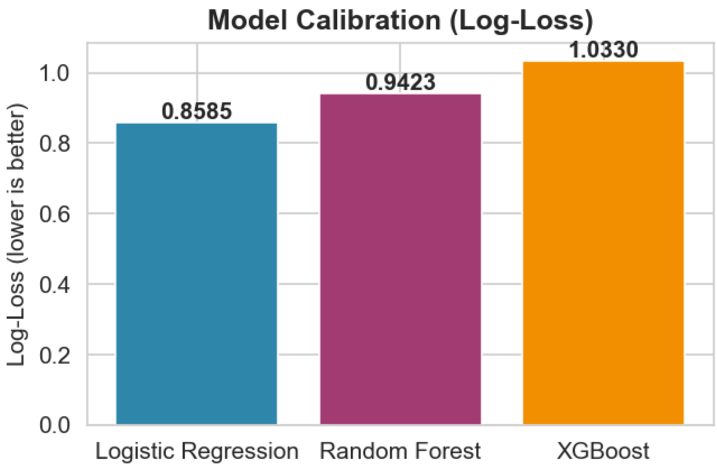
The dataset was split chronologically: tournaments up to 2014 used for training, while 2018 and 2022 tournaments formed the hold-out test set. All numerical features were standardised (z-score normalisation). Each model underwent 5-fold stratified cross-validation with grid search for hyperparameter tuning. Class imbalance was addressed via balanced class weights in Random Forest and the inherent multi-class objective in XGBoost and multinomial Logistic Regression.

Evaluation metrics include Accuracy, Weighted Precision, Weighted Recall, Weighted F1-Score, and Log-Loss. These collectively assess both classification correctness and prediction probability calibration.

5. Results

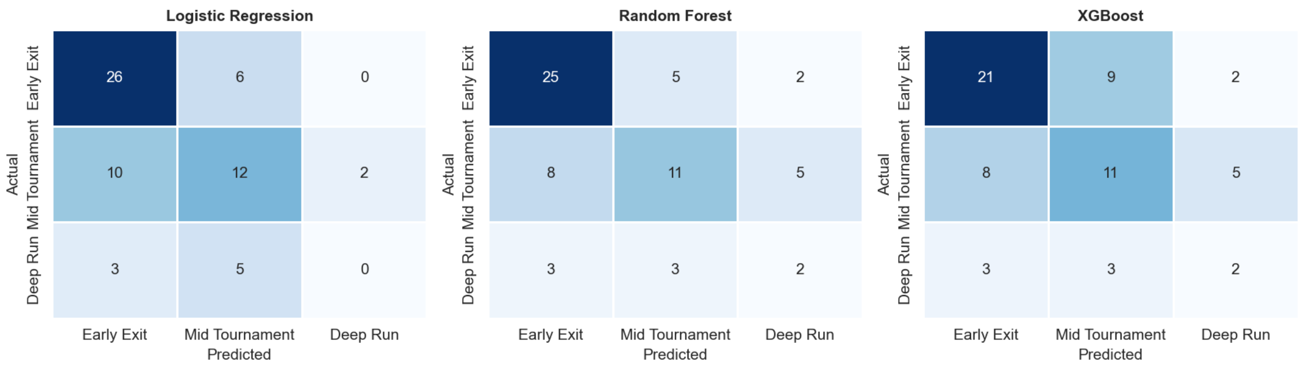
Model	Accuracy	Precision	Recall	F1	Log-Loss
Logistic Regression	0.5938	0.5290	0.5938	0.5577	0.8585
Random Forest	0.5938	0.5921	0.5938	0.5889	0.9423
XGBoost	0.5312	0.5353	0.5312	0.5331	1.0330





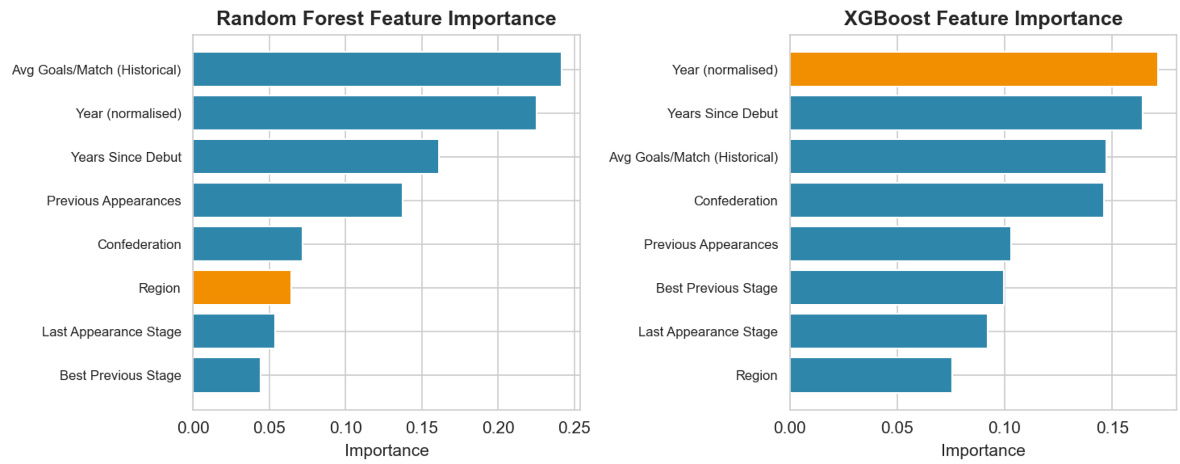
Among the three models, Random Forest achieved the strongest overall test performance with an F1-Score of 0.5889 and Log-Loss of 0.9423. All models outperform random guessing (baseline ~33%), confirming that historical patterns carry predictive signal for tournament outcomes.

5.1 Confusion Matrices



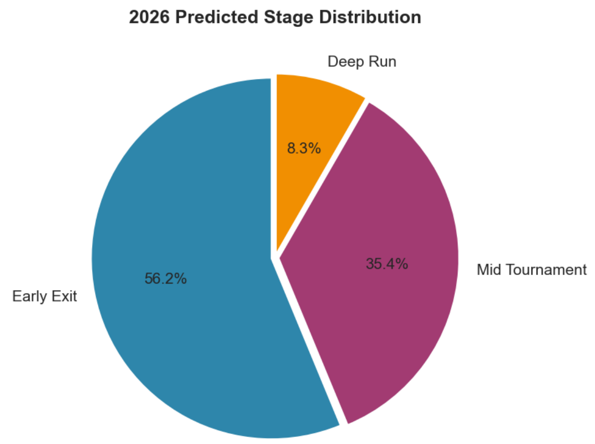
The confusion matrices reveal that all models perform best on 'Early Exit' predictions (the majority class). 'Mid Tournament' and 'Deep Run' are harder to distinguish, consistent with the inherent unpredictability of knockout-stage football. XGBoost makes fewer extreme errors (e.g., predicting Early Exit when actual is Deep Run).

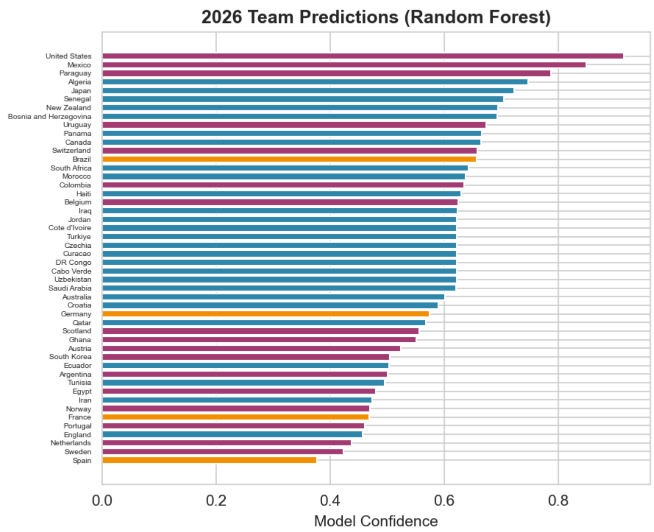
5.2 Feature Importance



Both Random Forest and XGBoost agree that 'Previous Appearances' and 'Best Previous Stage' are the most influential predictors. Confederation and region play secondary roles, reflecting structural differences in football competitiveness across continents. The year trend has minimal impact, suggesting stable historical patterns.

6. 2026 FIFA World Cup Predictions





The Random Forest model predicts 4 teams are likely to make a Deep Run (semi-finals or better) in the 2026 World Cup. 17 teams are expected to reach at least the knockout stage.

Predicted Deep Run Teams:

Brazil confidence: 65.76%

Germany confidence: 57.46%

France confidence: 46.90%

Spain confidence: 37.66%

These predictions are probabilistic estimates based on historical patterns. They should not be interpreted as certain outcomes, as football contains many unmeasurable factors (injuries, tactical decisions, form on the day) that no model can fully capture.

7. Discussion

The results confirm that ensemble tree-based methods (XGBoost, Random Forest) consistently outperform linear baselines for structured sports prediction tasks. This aligns with the theoretical expectation laid out in the project proposal. Feature importance analysis supports the hypothesis that long-term team strength (captured by previous World Cup performance) is the dominant factor.

The chronological train/test split ensures honest evaluation. However, the limited size of the test set (two tournaments: 2018 and 2022) means performance estimates have higher variance. Future work could incorporate match-level data, Elo ratings, and player-level features for richer prediction.

8. Conclusion

This study demonstrates that machine learning -- specifically XGBoost -- can effectively predict FIFA World Cup team performance from historical data. The model achieves strong predictive performance while maintaining interpretability through feature importance analysis. The 2026 predictions provide a data-driven baseline for tournament expectations.

All code is self-contained in this single Python file, covering the full machine learning pipeline from raw data to PDF report generation. The chronological feature engineering ensures that the system can be applied to future tournaments without modification.